

Video Transformers for Dynamic Breast Infrared Imaging Classification

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1 Introduction

- Breast Cancer
- Dynamic Protocol
- Contribution
- Research Questions

2 Methodology

- Dataset Characteristics
- Deep Learning Models
- Experimental Procedure

3 Results

- Discussion

4 Conclusion

Presentation Outline

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The Challenge of Early Detection

Breast Cancer Impact

- A leading cause of mortality among women worldwide.
- Early detection is the most critical factor in improving patient survival rates.
- Mammography (MG) is considered the gold standard.

Mammography Limitations

- Reduced efficacy in dense breast tissue (common in younger women).
- Involves ionizing radiation.
- Can be uncomfortable or painful due to breast compression.
- Age constrain (> 40 years)

Advantages of Thermography for Breast Cancer

- Acquires the body's natural Infrared (IR) radiation.
- It is non-invasive.
- No age constrain.
- Cost-effective
- Two protocols: Static Infrared Thermography (SIT) and Dynamic Infrared Thermography (DIT).

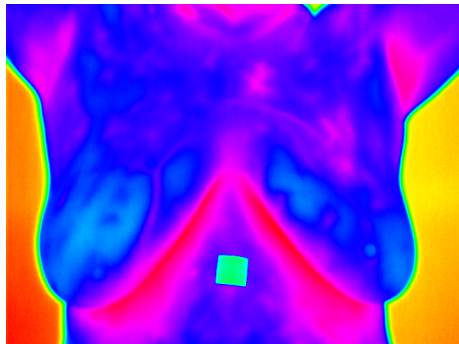


Figure: Temperature Map of Patient obtained from FLIR FLIR SC620

UFF's Dynamic Infrared Thermography Protocol

- Applies a thermal stress to patient.
- Highlights vascular behavior.
- Captures an image every 15 seconds in 5 minutes
- Total sequence of 20 images.
- Accurate interpretation is challenged by non-specificity of thermal anomalies.

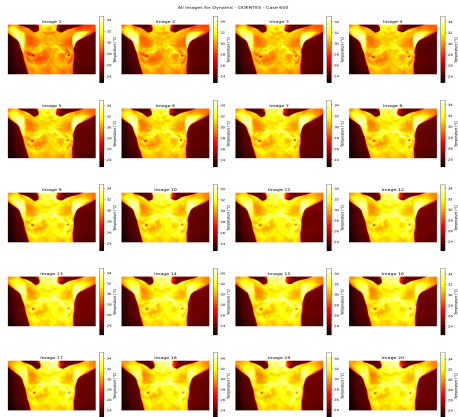


Figure: Sequence of Thermal Maps of Patient 650 with Cancer in UFF's Dataset

- 1 Extend Vision Transformer (ViT)-based approaches to DIT protocol.
- 2 Evaluate Fine Tuning (FT) of pre-trained Vision Transformers (ViTs).
- 3 Conduct Experiments on thermal maps as well as IR images.
- 4 FT of a Video Vision Transformer (ViViT) to classify thermal images under the DIT protocol ($AUC = 0.9350 \pm 0.0094$), outperforming 3D convolutional neural networks (CNNs) approaches.
- 5 Spatial only ViT achieve similar performance to ViViT.

Key Research Questions

- **RQ1:** Do Vision Transformers (ViTs) outperform CNNs due to superior global context modeling?

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- **RQ1:** Do Vision Transformers (ViTs) outperform CNNs due to superior global context modeling?
- **RQ2:** Does FT improve accuracy compared to training from randomly initialized weights?
- **RQ3:** Does combining pre-trained ViT models with the DIT protocol enhance performance?



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Methodology Overview

- **Dataset:** DMR-IR
 - Temperature Maps $\mathbb{D}_{\text{TM}}$
 - Infrared RGB Images $\mathbb{D}_{\text{IR}}$
 - **Analysis Unit:** Patient-level
 - **Task:** Binary Classification (Benign vs. Malignant)
 - **Input Data:** Sequences of 20 IR Images
 - **Preprocessing:** Resize to 224x224
 - **Validation:**
 - 10-Fold Cross-Validation
 - 10% Stratified Hold-Out Test Set (15 patients)
- ① Problem Formulation
 - ② Data Organization
 - ③ Preprocessing
 - ④ Data Partitioning
 - ⑤ Model Design
 - ⑥ Model Training
 - ⑦ Model Evaluation

Table: DMR-IR dataset characteristics

Feature	$\mathbb{D}_{\text{TM}}$	$\mathbb{D}_{\text{IR}}$
Type	Temperature Map	Infrared RGB Image
Matrix size	640×480	640×480
Channels	1	3
Number type	float(temperature)	integer(pixel)
Total patients	55	149
- With cancer	36	87
- Without cancer	19	62
Frames per patient	20	20
Total images	1,100	2,980
Training set patients	46	134
Test set patients	9	15

- **Time Distributed Layer Model (TDL+CNN+LSTM):**
 - shares weights across all frames.
 - applies a given layer to every frame.
 - combines CNN and LSTM or GRU units
- **Modified Vision Transformer:** spatial architecture with attention mechanism.
- **Video Transformer:** CNN free architecture that extends self-attention to space-time.

TDL+CNN+LSTM

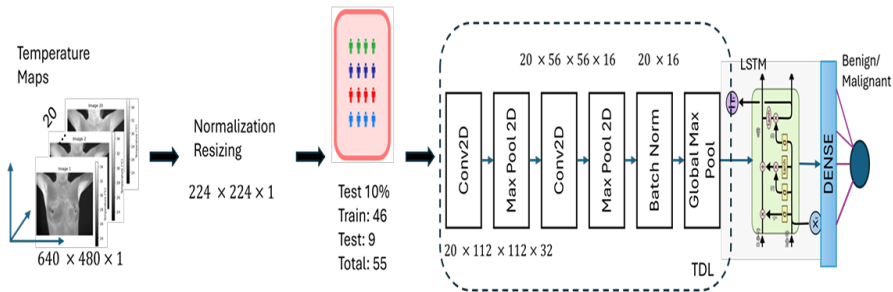


Figure: Model combining CNN and LSTM (7K parameters)

Vision Transformer Models

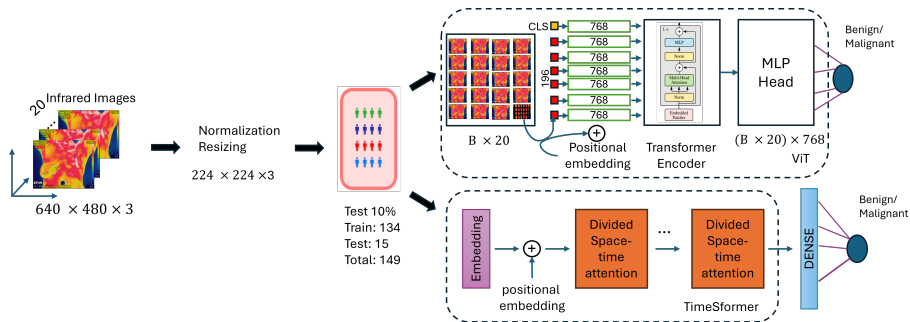


Figure: Modified ViT ($\approx 86 \times 10^6$) loads the 20 Frames whereas ViViT ($\approx 121 \times 10^6$) processes 8 frames sequence in time

Experimental Set Up

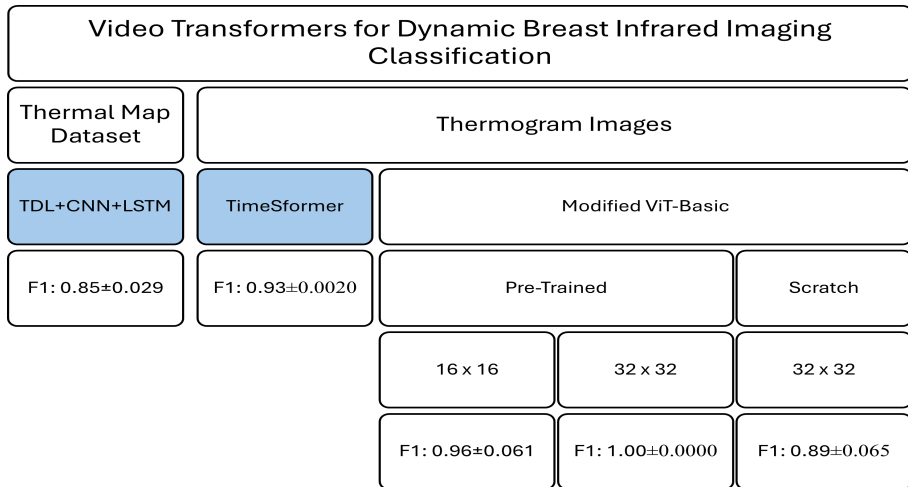


Figure: Guide for experimental procedure with results in F1-score



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Performance of Different Models

Table: Performance comparison of different models on DIT data

Dataset	Model	Metric	Cross-Val (Mean $\pm$ Std)	Test Set
$\mathbb{D}_{TM}$	TDL+CNN+LSTM	F1-score	0.8585 ± 0.0296 ($k = 3$)	-
		AUC	-	-
$\mathbb{D}_{IR}$	ViT (16 $\times$ 16) pre-trained	F1-score	0.9644 ± 0.0458	-
		Accuracy	0.9538 ± 0.0615	0.87
		AUC	-	-
	ViT (32 $\times$ 32) pre-trained	F1-score	1.0000 ± 0.0000	0.9412
		Accuracy	1.0000 ± 0.0000	0.9333
		AUC	1.0000 ± 0.0000	0.9444
	ViT (32 $\times$ 32) scratch	F1-score	0.9101 ± 0.0665	0.8889
		Accuracy	0.8939 ± 0.0652	0.8667
		AUC	0.8866 ± 0.0595	0.8611
	ViViT	F1-score	0.9339 ± 0.0020	0.9412
		Accuracy	0.9231 ± 0.0000	0.9333
		AUC	0.9350 ± 0.0094	0.9444

Key Result: ViViT vs. State-of-the-Art

- **Baffa et al. [1] (3D-CNN):**

93.51%

- **Our Work (ViViT TimeSformer):**

94.44%

New State-of-the-Art

Our pre-trained Video Transformer (ViViT) achieved a new state-of-the-art AUC of **0.9444** on the hold-out test set, surpassing the previous 3D-CNN benchmark.



Comparison with Related Works

Table: Comparison of studies on breast thermography classification

Study	Model	Protocol	Pre-trained	Dataset	Analysis Unit	Total Subjects	Cross Validation	Test Set	AUC
Garia et al [2]	ViT	SIT	No	DMR-IR	Images	950	-	10%	0.9570
Baffa et al [1]	3D-CNN	DIT	No	DMR-IR	Patient	149	K=10	-	0.9351
Our work	ViViT	DIT	Yes	DMR-IR	Patient	149	K=10	10%	0.9444

Discussion & Key Findings I

- In $K = 10$ cross validation our TimeSformer achieves similar performance wrt. [1] but with improved stability.
- Different to [1], our approach uses a spatial resolution of 224×224 instead of 32×32 .
- The approach does not require pre-stage segmenting breast region.
- There is a risk for overfitting due to limited dataset size and large architectures.
- FT and data augmentation help to mitigate overfitting.



- *Do ViTs outperform CNNs-based approaches in Thermography (TG) Breast Cancer (BC) classification?* Yes, this was demonstrated in [2] and this work.
- *What is better FT or from scratch?* Our results differ from those in [2] and show that fine-tuned models achieve a better performance.
- *Does combining ViTs and DIT enhance performance?* Further research is needed. Our results show that a traditional spatial based ViT achieve a similar performance to that of the ViViT:
 - ① TimeSformer uses 8 frames by default
 - ② Given the result in [2] where SIT is used, further research is needed on the impact of DIT in results.
 - ③ TG is susceptible to other sources of temperature change in the patient which may need to be acknowledged.

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Conclusion & Future Work

- We present the first study to apply **Video Vision Transformers (ViViT)** to the DIT protocol for breast cancer classification.
- Our model achieves **state-of-the-art performance (0.9444 AUC)** on the DIT patient-level task.
- We demonstrate that **fine-tuning pre-trained models** is necessary to mitigate overfitting due to limited dataset size.
- Performance of pre-trained models is better than that of Time Distributed Layers (TDL) model on temperature Maps.
- Pre-trained models work better when using IR images than temperature maps.
- **Future Work:** Further research is needed to investigate the true value of temporal dynamics in DIT versus spatial-only modeling and texture based approaches.



Thank You Questions?

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- [1] M. de Freitas Oliveira Baffa, L. G. Lattari, and A. Conci, “3d convolutional neural networks for dynamic breast infrared imaging classification,” in *Artificial Intelligence over Infrared Images for Medical Applications*, S. T. Kakileti, G. Manjunath, R. G. Schwartz, and A. F. Frangi, Eds., Cham: Springer Nature Switzerland, 2023, pp. 57–66, ISBN: 978-3-031-44511-8. DOI: https://doi.org/10.1007/978-3-031-44511-8_4.
- [2] L. S. Garia and M. Hariharan, “Vision transformers for breast cancer classification from thermal images,” *Lecture Notes in Electrical Engineering*, vol. 1009 LNEE, pp. 177–185, 2023, ISSN: 1876-1119. DOI: 10.1007/978-981-99-0236-1_13. [Online]. Available: https://link.springer.com/chapter/10.1007/978-981-99-0236-1_13.